

Ryan Camilleri

Observational cosmology

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Summary

I am a postdoctoral research fellow at The University of Queensland specialising in observational cosmology, with a focus on standard candle techniques using Type Ia supernovae and tests of beyond- Λ CDM models. I have strong experience building end-to-end cosmology pipelines for data analysis, simulation, and parameter inference alongside statistical analysis and model testing. I am also proficient working with established cosmological software frameworks such as COSMOSIS, SNANA, and COBAYA.

Education

The University of Queensland, Doctor of Philosophy in Supernova Cosmology with the Dark Energy Survey – Brisbane, Australia June 2022 – Sept 2026

The University of Queensland, Bachelor of Science (Honours) in Physics – Brisbane, Australia Feb 2021 – June 2022

Griffith University, Bachelor of Science in Physics & Mathematics – Gold Coast, Australia Feb 2017 – Dec 2020

Awards

Honourable Mention for Best Student Publication in Physics 2025
Awarded at The University of Queensland

Dark Energy Survey “Builder” 2025
Recognition for contributions to the DES Collaboration

Best Presentation - non-invited Speaker 2024
Awarded at the Frontiers and Gravitational physics conference in Portsmouth, England.

Head of School Award for Excellence in Science 2020
Awarded to the Griffith University student in their final year with the highest career cumulative GPA in the Bachelor of Science (GPA 7.0).

Australian Institute of Physics Prize 2020
Awarded to the most outstanding graduating student in a Physics major (GPA 7.0).

Griffith Futures Scholarship 2020
Awarded to students who have been impacted by various types of educational disadvantage.

Griffith Award for Academic Excellence 2017
Awarded to students who achieve a cumulative GPA of at least 6.0, calculated over the academic year. I received this award each year of my undergraduate degree.

Publications

Supernova Unite: Constraints on cosmologies beyond- Λ CDM 2026
In preparation
R. Camilleri et al.

Supernova Unite: Combining Pantheon+ and DES-SN5YR R. Camilleri et al. (Submitted to Publications of the Astronomical Society of Australia)	2026
Supernova Unite: Host-Galaxy Mass Measurements of Type Ia Supernovae and Their Impact on Cosmology J. lee, R. Camilleri et al. (Submitted to Astrophysical Journal)	2026
DES Supernova Program: H_0 from the Inverse Distance Ladder R. Camilleri et al. (Monthly Notices of the Royal Astronomical Society)	2025
DES Supernova Program: Investigating Beyond-ΛCDM R. Camilleri et al. (Monthly Notices of the Royal Astronomical Society)	2024
DES Supernova Program: Reanalysis of Cosmology Results B. Popovic et al. (Monthly Notices of the Royal Astronomical Society)	2025
Dark Energy Survey: implications for cosmological expansion models from the final DES Baryon Acoustic Oscillation and Supernova data DES Collaboration (Physical Review D)	2025
Cosmological non-linearities as an effective fluid L. Giani, R. v. Marttens, R. Camilleri (Physical Review Letters)	2025
WiFeS observations of nearby southern Type Ia supernova host galaxies A. Carr, T. M. Davis, R. Camilleri et al. (Publications of the Astronomical Society of Australia)	2024

Activities

UQ-QASMT Research Mentoring Program I guided a group of Year 10 students through an independent cosmology project, introducing them to Python coding, and helping them develop the skills to produce clear plots and reports that effectively communicate their results.	July 2025 – 2026
Dark Energy Survey I organised and led tri-weekly international meetings, facilitating collaboration between Australia and UK teams.	2022 – 2025
UQ Astro Work Experience Program The Astronomy group at The University of Queensland runs a weeklong work experience program (typically mid-year) and offers future STEM students who want careers in astronomy a first-hand look at the day-to-day life of an Astronomer. The week consists of an astronomy project, telescope demonstrations, “Meet the researcher” discussions and on the final day, the students get to present what they have learned to the group. Since 2022, I have volunteered to help organise and run this program.	2022 – present

Conferences

Next Generation of SN Surveys Mini-Workshop – Oxford, United Kingdom Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR	July 2026
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Supernova Cosmology: New Challenges and Opportunities – Seoul, South Korea

June 2026

Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR

The End of Λ ? – Incheon, South Korea

Sept 2025

Invited talk presenting both the DES supernova cosmology results and my current project, which is a combined DES & Pantheon+ analysis

Cosmic Flows – Brisbane, Australia

Feb 2025

Presented H_0 inverse distance ladder results

DES Spring Meeting – Barcelona, Spain

May 2024

Presented beyond- Λ CDM paper

Frontiers in Cosmology & Gravitational Physics – Portsmouth, England

May 2024

Presented beyond- Λ CDM paper

OzGrav ECR Winter School – Kingscliff, Australia

Aug 2025

Attended the OzGrav ECR Winter School

Memberships

Dark Energy Survey (DES) Collaboration: Builder. The Dark Energy Survey is an international, collaborative effort to map hundreds of millions of galaxies, detect thousands of supernovae, and find patterns of cosmic structure that will reveal the nature of the mysterious dark energy that is accelerating the expansion of our Universe.

ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav): Member. The mission of OzGrav is to capitalise on the historic first detections of gravitational waves to understand the extreme physics of black holes and warped spacetime, and to inspire the next generation of Australian scientists and engineers through this new window on the Universe.

Dark Energy Spectroscopic Instrument (DESI) Collaboration: Member. DESI seeks to map the large-scale structure of the Universe over an enormous volume. Targeting about 30 million pre-selected galaxies across one-third of the night sky, scientists will use DESI's redshifts data to construct 3D maps of the universe.

References

Prof. Tamara Davis

PhD Supervisor, University of Queensland — tamarad@physics.uq.edu.au

A/Prof. Maria Vincenzi

DES Supernova Working Group Convenor — maria.vincenzi@physics.ox.ac.uk

Dr Paul Shah

DES Collaboration — paul.shah.19@ucl.ac.uk